Document Information

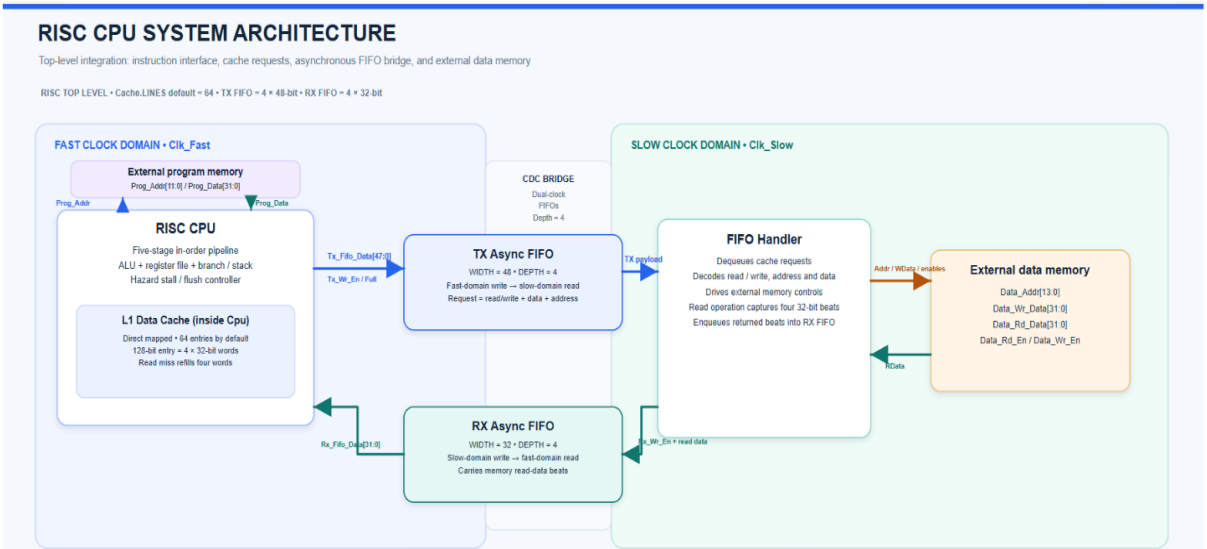
Field	Detail
Project Name	**CPU Lite**
Top-Level Module	Risc
Project Engineer	Sethumadhavan Bharathi
Version	V1.0
Date	12-October-2026

Introduction:

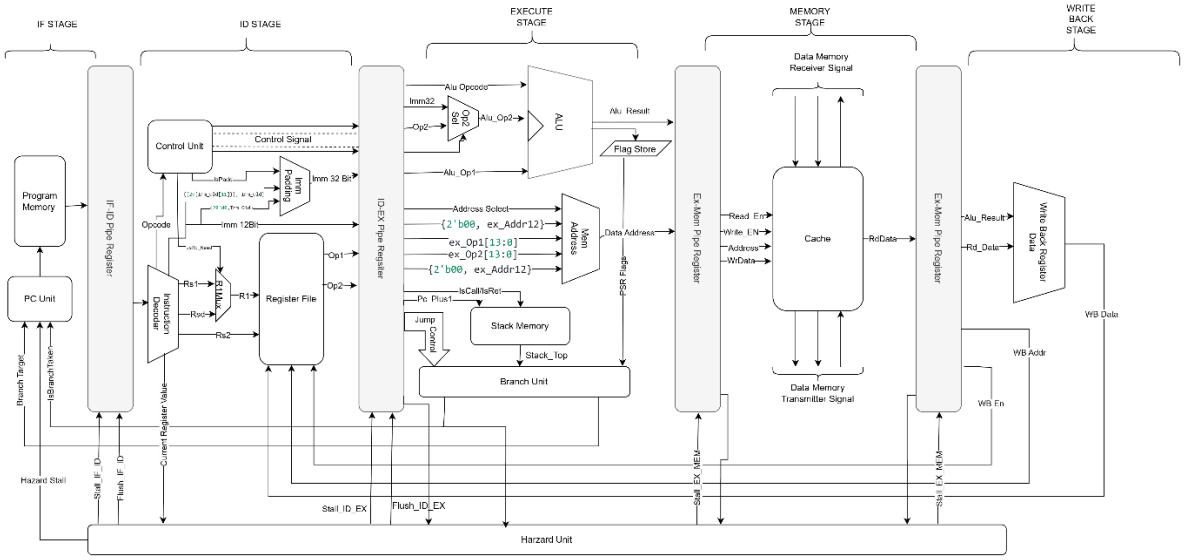
This project focuses on designing a 32-bit, single-issue, in-order RISC CPU Lite using a five-stage pipeline consisting of Instruction Fetch, Instruction Decode, Execute, Memory Access, and Write Back. The processor supports arithmetic, logical, memory access, branch, call, and return operations, with an eight-entry stack for handling procedure calls and returns. Pipeline stalls and flushes are used to manage data hazards and changes in instruction flow. The CPU and its L1 data cache operate at a reference frequency of 150 MHz, while the data memory operates at 70 MHz. The L1 data cache handles LOAD and STORE operations, and cache misses are serviced by the data memory through a dedicated CDC bridge that enables safe communication between the asynchronous clock domains. The design also includes separate TX and RX FIFOs, each with a depth of four entries, for data buffering and transfer.

Architecture:

System Level:



CPU Level:



5-Stage Operation:

1.Instruction Fetch (IF) Stage:

- The Program Counter (PC) provides the address of the instruction to be fetched from the read-only program memory.
- During reset, the PC is initialized to address zero.
- Under normal operation, the PC increments by one to fetch the next instruction.
- When a branch is taken, the PC is updated with the target address to fetch the instruction from the new location.
- The PC is held when a memory stall occurs or a hazard stall is active, unless a branch is taken.
- When a halt signal is received, the processor stops updating the PC and remains halted until reset
- Module Involved: PC Unit

2.Instruction Decode (ID)Stage:

- The Instruction Decoder identifies the operation to be performed by interpreting the fetched instruction.
- The Control Unit generates the necessary control signals based on the instruction type.
- The Register File reads the source operands required for instruction execution.
- The Immediate Padding unit extends the immediate value to 32 bits when required by the instruction.
- The decoded instruction, operands, immediate value, and control signals are transferred to the Execute stage through the ID-EX pipeline register.

- Module Involved: Control Unit , Register File, Instruction Decoder , Imm Padding

3.Execute (EX) Stage:

- The ALU performs arithmetic and logical operations using the operands received from the Instruction Decode stage.
- The Operand 2 selection unit selects either a register operand or an immediate value as the second ALU input.
- The ALU generates the result and updates the processor flags when required.
- The Branch Unit evaluates branch conditions and determines the target address for branch, jump, call, and return operations.
- The Stack Memory supports procedure calls and returns using the eight-entry stack.
- The execution result, memory access information, and relevant control signals are transferred to the Memory stage through the EX-MEM pipeline register.
- Module Involved: ALU Unit , Flag Store ,Branch Unit , Stack Memory , Memory Address Mux .

4.Memory Access (MEM) Stage:

- The L1 data cache is the first point of access for all LOAD and STORE operations.
- For a cache hit, the required data is accessed directly through the cache.
- For a cache miss, the memory request is sent to the data memory through the CDC bridge.
- The CDC bridge manages communication between the CPU/cache clock domain at 150 MHz and the data memory clock domain at 70 MHz.
- The TX and RX FIFOs, each with a depth of four entries, support data transfer and buffering.
- When a cache miss requires a memory response, the pipeline is held as necessary until the required operation can proceed. Returned data is used to update the cache and provide the requested LOAD data.

5.Write Back (WB) Stage:

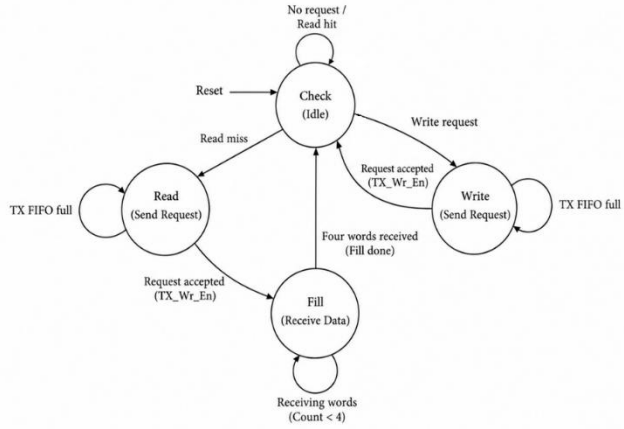
- The Write Back stage selects the appropriate result to be written into the Register File.
- For arithmetic and logical instructions, the result is obtained from the ALU.
- For LOAD instructions, the result is obtained from the data returned by the memory access path.
- The Write Back register holds the required result and associated destination information.

- The destination register is updated when the write enable signal is active, allowing subsequent instructions to use the updated value.

Cache:

- The design uses a parameterized, direct-mapped L1 data cache with 64 lines by default.
- Each cache line stores four 32-bit words, providing a default capacity of 1 KB.
- The cache uses the address tag, index, and word offset to locate the requested data.
- On a read hit, data is returned directly; on a miss, four words are fetched from memory to fill the cache line.
- The cache uses a write-through policy, forwarding write requests to memory and updating the cache on a write hit.
- It uses no-write-allocate, meaning a write miss is sent to memory without allocating a new cache line.

The cache controller uses four FSM states: Check, Read, Fill, and Write.



Stack:

- The Stack module is used to store return addresses during procedure calls and retrieve them during return operations.
- It contains eight entries, with each entry storing a 12-bit address.
- During a push operation, the input address (Din) is stored at the current stack position, and the stack pointer is incremented.
- During a pop operation, the stack pointer is decremented to access the previous address.

- The Top output provides the most recently stored address. When the stack is empty, Top returns zero.
- The stack prevents push operations when full and pop operations when empty.
- An active-low reset clears the stack pointer and initializes all stack entries to zero.

Hazard Unit:

- The Hazard Unit detects data dependencies between the current instruction and instructions in the ID/EX, EX/MEM, and MEM/WB pipeline stages.
- It compares the source register addresses of the current instruction with destination register addresses from previous instructions.
- Register usage control signals determine which source registers must be checked for dependencies.
- When a dependency is detected, the Hazard_Stall signal is asserted to prevent the processor from using an incorrect or unavailable result.
- The unit generates stall signals to hold the IF/ID stage during a hazard and holds the pipeline stages when a memory stall occurs.
- Flush signals clear instructions when a branch is taken or a halt occurs. The ID/EX stage is also flushed when a data hazard is detected.
- These stall and flush mechanisms help maintain correct instruction execution throughout the pipeline.
- Module Involved: Hazard_Unit

Data Path Signal:

Pipe Register	**Carries**
IF/ID	Instruction[31:0], Pc_Plus1[11:0]
ID/EX	Op1[31:0], Op2[31:0], Imm_Ext[31:0], Rs1[3:0] (add for forwarding), Rs2[3:0] (add for forwarding), Rd[3:0], Addr12[11:0], Pc_Plus1[11:0], Mem_Addr[1:0], Alu_Op[4:0], Alu_Src, Flag_Store[3:0], Branch, Br_Cond[2:0], Jump, Jump_Reg, Jump_If, Is_Call, Is_Ret, Halt, Mem_Rd, Mem_Wr, Reg_Wr, Wb_Sel
EX/MEM	Alu_Result[31:0], Store_Data[31:0], Mem_Address[13:0], Rd[3:0], Mem_Rd, Mem_Wr, Reg_Wr, Wb_Sel
MEM/WB	Alu_Result[31:0], Rd_Data[31:0], Wb_Sel, Rd[3:0], Reg_Wr

Stage	Units	Inputs	Outputs
IF	Pc_Unit, program memory	Target, isBranchTaken, stalls	Instruction, Pc_Plus1
ID	Instruction_Decode, Control_Unit, Imm_padding, Register_File	Instruction, wb_Wb_Data, wb_Rd	Op1, Op2, Imm_Ext, control signals
EX	Alu_Unit, Flag_Store, Branch_Unit, Stack	Op1, Op2 or immediate, flags	Alu_Result, Target, isBranchTaken
MEM	Cache (L1)	Mem_Address, Store_Data, Mem_Rd, Mem_Wr	Rd_Data, Done
WB	Wb_Data_Sel mux, register file write port	Alu_Result, Rd_Data, Wb_Sel	Wb_Data to the register file

Control Path Signals:

Signal	Meaning
Alu_Op[4:0]	Which ALU operation
isImm	ALU operand B is the immediate instead of Rs2
Imm_pad	1 = sign-extend the 12-bit immediate, 0 = zero-extend
Wb_En	Write the result to the register file
Wb_Sel	0 = ALU result, 1 = data from the cache
Mem_Read / Mem_Write	Cache read / write
Mem_Addr[1:0]	Address source: 0 = 12-bit address field, 1 = Rs1 register, 2 = Rs2 register
isRd_Need	Read Rd as a source register (store data)
Flag_Store[3:0]	Which flags this instruction updates
Jump / Jump_if / Jump_Reg	JMP / conditional on Rs1 ≠ 0 / jump to Rs1
Branch, Br_Cond[2:0]	Conditional branch and its condition
isCall / isRet / Halt	Call, return, stop
Use_Rs1 / Use_Rs2	Tell the hazard unit which sources this instruction reads

Control Signals per instruction:

Instruction (Opcode)	Alu_Op	isImm	Imm_pad	Wb_En	Wb_Sel	Memory	Flag_Store (V,C,N,Z)	Uses
NOP (00)	0	0	0	0	0	0	0000	0
LOAD (01)	0	0	0	1	1	Read, Mem_Addr=0	0000	0
LOAD_IND (02)	0	0	0	1	1	Read, Mem_Addr=1	0000	Rs1
LOAD_IMM (03)	17	1	0	1	0	0	0000	0
STORE (04)	0	0	0	0	0	Write, Mem_Addr=0	0000	isRd_Need
STORE_IND (05)	0	0	0	0	0	Write, Mem_Addr=2	0000	isRd_Need, Rs2
ADD (06)	0	0	0	1	0	0	1111	Rs1, Rs2
SUB (07)	1	0	0	1	0	0	1111	Rs1, Rs2
MUL (08)	2	0	0	1	0	0	0011	Rs1, Rs2
AND (09)	3	0	0	1	0	0	0011	Rs1, Rs2
OR (0A)	4	0	0	1	0	0	0011	Rs1, Rs2
NOT (0B)	5	0	0	1	0	0	0011	Rs1
CMP (0C)	1	0	0	0	0	0	1111	Rs1, Rs2
EQ (0D)	7	0	0	1	0	0	0001	Rs1, Rs2
JMP (0E)	0	0	0	0	0	0	0000	Jump
JMP_IF (0F)	0	0	0	0	0	0	0000	Jump_if, Rs1
XOR (10)	6	0	0	1	0	0	0011	Rs1, Rs2

SHL (11)	10	0	0	1	0	0	0011	Rs1, Rs2
SHR (12)	11	0	0	1	0	0	0011	Rs1, Rs2
SAR (13)	12	0	0	1	0	0	0011	Rs1, Rs2
ROL (14)	13	0	0	1	0	0	0011	Rs1, Rs2
ROR (15)	14	0	0	1	0	0	0011	Rs1, Rs2
ADDI (16)	0	1	1	1	0	0	1111	Rs1
SUBI (17)	1	1	1	1	0	0	1111	Rs1
ANDI (18)	3	1	0	1	0	0	0011	Rs1
ORI (19)	4	1	0	1	0	0	0011	Rs1
XORI (1A)	6	1	0	1	0	0	0011	Rs1
MOV (1B)	15	0	0	1	0	0	0000	Rs1
SLT (1C)	8	0	0	1	0	0	0001	Rs1, Rs2
SLTU (1D)	9	0	0	1	0	0	0001	Rs1, Rs2
LUI (1E)	16	1	0	1	0	0	0000	0
BEQ (20)	0	0	0	0	0	0	0000	Branch, Br_Cond =0
BNE (21)	0	0	0	0	0	0	0000	Branch, Br_Cond =1
BLT (22)	0	0	0	0	0	0	0000	Branch, Br_Cond =2
BGE (23)	0	0	0	0	0	0	0000	Branch, Br_Cond =3

BLTU (24)	0	0	0	0	0	0	0000	Branch, Br_Cond =4
BGEU (25)	0	0	0	0	0	0	0000	Branch, Br_Cond =5
JMP_REG (26)	0	0	0	0	0	0	0000	Jump_Reg, Rs1
CALL (27)	0	0	0	0	0	0	0000	isCall
RET (28)	0	0	0	0	0	0	0000	isRet
HALT (FF)	0	0	0	0	0	0	0000	Halt

ALU operation codes:

Code	Op	Code	Op	Code	Op
0	ADD	6	XOR	12	SAR
1	SUB (also CMP)	7	EQ	13	ROL
2	MUL	8	SLT	14	ROR
3	AND	9	SLTU	15	MOV
4	OR	10	SHL	16	LUI
5	NOT	11	SHR	17	LOAD_IMM

Instruction Set:

Instruction Format:

Every instruction is a fixed 32-bit word. Field positions are constant; not every field is used by every instruction.

Field	Bits	Width	Meaning
OPCODE	[31:24]	8	Operation code.
Rd / OUT	[23:20]	4	Destination register (also the data-source register for STORE).

Rs1 / OP1	[19:16]	4	First source register; base register for register-indirect addressing.
Rs2 / OP2	[15:12]	4	Second source register; shift amount uses Rs2[4:0].
IMM / ADDR	[11:0]	12	12-bit immediate or direct address.

- Direct LOAD/STORE use the 12-bit ADDR and reach the first 4K words of data memory. Register-indirect LOAD_IND/STORE_IND use a register as a full 14-bit address and reach the entire 16K.
- LOAD_IMM zero-extends the immediate. ADDI/SUBI sign-extend it. ANDI/ORI/XORI zero-extend it. Shifts take their amount from Rs2[4:0].

Core Instruction:

No Operation (Stall)

Mnemonic	Opcode	Fields used	Operation
NOP	00	—	No operation.

Load/Store Instructions

Mnemonic	Opcode	Fields used	Operation
LOAD	01	Rd, ADDR	Rd ← DATA[ADDR] (via cache)
LOAD_IND	02	Rd, Rs1	Rd ← DATA[Rs1] (via cache)
STORE	04	Rd(src), ADDR	DATA[ADDR] ← Rd (through cache)
STORE_IND	05	Rd(src), Rs2	DATA[Rs2] ← Rd

Register type Instructions

Mnemonic	Opcode	Fields used	Operation
ADD	06	Rd, Rs1, Rs2	Rd ← Rs1 + Rs2
SUB	07	Rd, Rs1, Rs2	Rd ← Rs1 – Rs2
MUL	08	Rd, Rs1, Rs2	Rd ← (Rs1 × Rs2)[31:0]
AND	09	Rd, Rs1, Rs2	Rd ← Rs1 & Rs2
OR	0A	Rd, Rs1, Rs2	Rd ← Rs1 | Rs2
NOT	0B	Rd, Rs1	Rd ← ~Rs1

XOR	10	Rd, Rs1, Rs2	Rd ← Rs1 ^ Rs2
CMP	0C	Rs1, Rs2	flags ← (Rs1 – Rs2); no write-back
EQ	0D	Rd, Rs1, Rs2	Rd ← (Rs1 == Rs2) ? 1 : 0
ADDI	16	Rd, Rs1, IMM	Rd ← Rs1 + sign_ext(IMM)
SUBI	17	Rd, Rs1, IMM	Rd ← Rs1 – sign_ext(IMM)
SHL	11	Rd, Rs1, Rs2	Rd ← Rs1 << Rs2[4:0] (logical)
SHR	12	Rd, Rs1, Rs2	Rd ← Rs1 >> Rs2[4:0] (logical)
SAR	13	Rd, Rs1, Rs2	Rd ← Rs1 >>> Rs2[4:0] (arithmetic)
ROL / ROR	14 / 15	Rd, Rs1, Rs2	Rotate left / right by Rs2[4:0]
ANDI / ORI / XORI	18 / 19 / 1A	Rd, Rs1, IMM	Logical op with zero_ext(IMM)
MOV	1B	Rd, Rs1	Rd ← Rs1
SLT / SLTU	1C / 1D	Rd, Rs1, Rs2	Rd ← (Rs1 < Rs2) ? 1 : 0 (signed / unsigned)
LUI	1E	Rd, IMM	Rd ← {IMM, 20’b0}
LOAD_IMM	03	Rd, IMM	Rd ← zero_ext(IMM)

Branch Instructions

Mnemonic	**Opcode**	**Fields used**	**Operation**
JMP	0E	ADDR	PC ← ADDR
JMP_IF	0F	Rs1, ADDR	if (Rs1 ≠ 0) PC ← ADDR
BEQ	20	ADDR	if Z: PC ← ADDR
BNE	21	ADDR	if ¬Z: PC ← ADDR
BLT	22	ADDR	if (N ⊕ V): PC ← ADDR (signed <)
BGE	23	ADDR	if ¬(N ⊕ V): PC ← ADDR (signed ≥)

BLTU / BGEU	24 / 25	ADDR	Unsigned < / ≥ branch (uses if C = 0, BLTU valid | if C = 1, BGEU valid)
JMP_REG	26	Rs1	PC ← Rs1[11:0] (computed jump / return)
CALL / RET	27 / 28	ADDR / —	Subroutine call / return (define a link register + stack)

Stop

Mnemonic	**Opcode**	**Fields used**	**Operation**
HALT	FF	—	Stop execution.

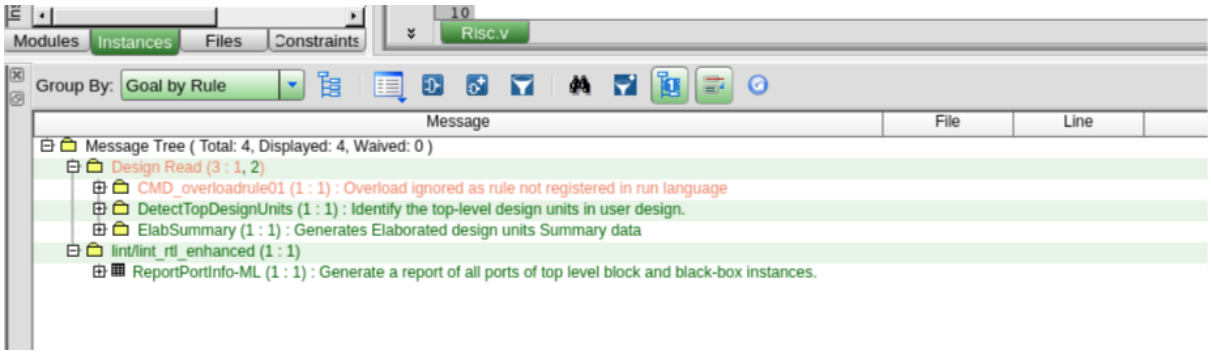
STATIC ANALYSIS:

LINT:

Lint/rtl

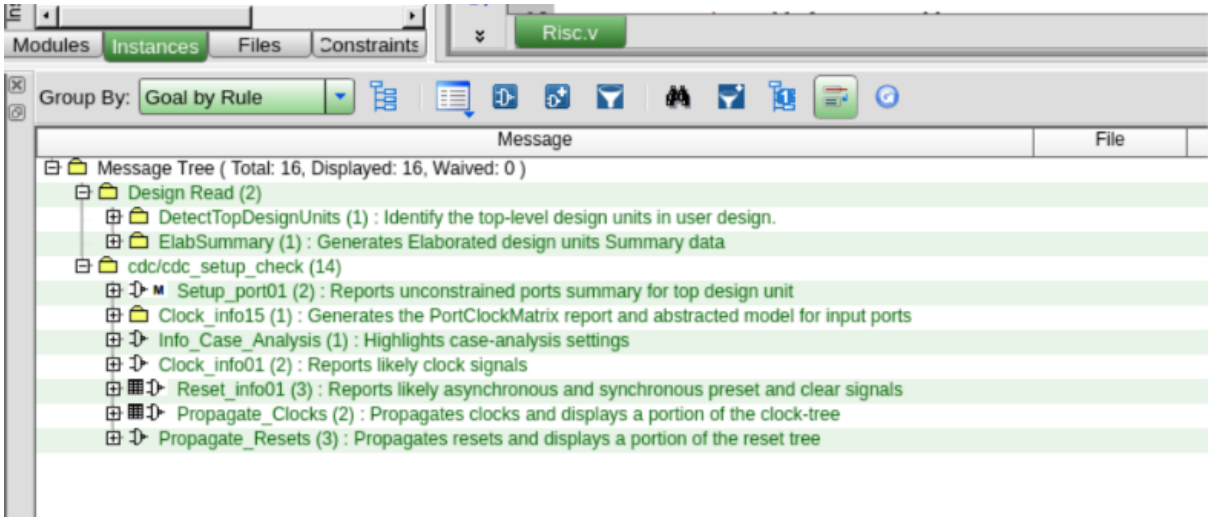


Lint/rtl_enchanced

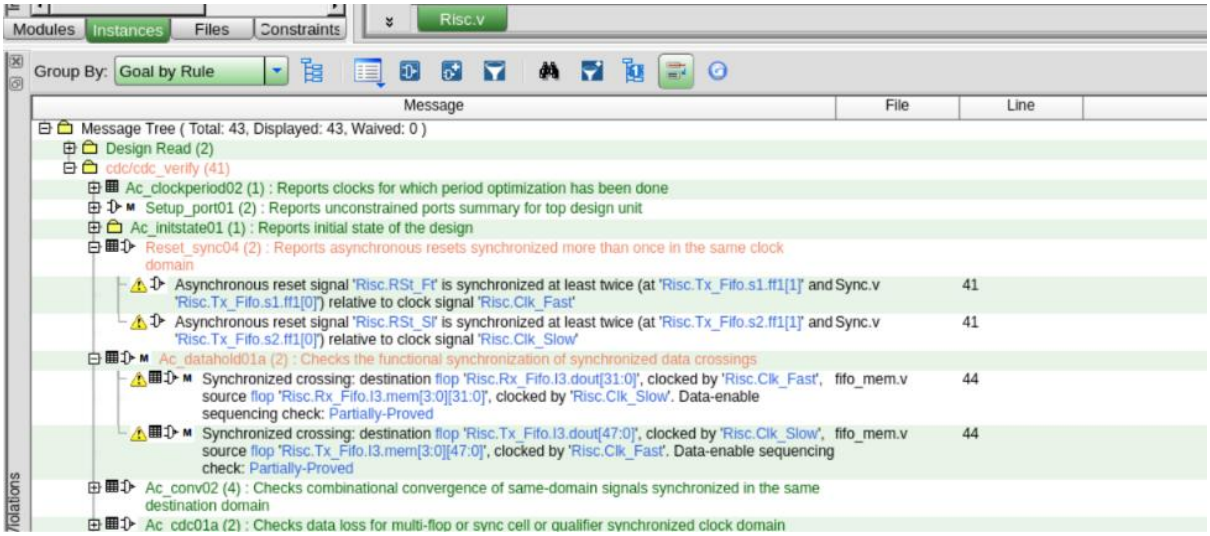


CDC:

Cdc/setupcheck



CDC Verify



Warning:

1- **Reset_sync04** - Sync and Reset_Sync module are the same two-flop shape, so the tool reads the Gray-pointer synchroniser as a second reset synchroniser. Sync.q drives the pointer comparator, never a reset pin. There is one Reset_Sync per domain, so one reset-release time per domain**.**

2 – **Ac_datahold01a** - Async FIFO memory crossing. Data words are stable because wr_ptr advances only after the write and crosses as Gray code, so Empty releases the reader at least two clocks later. “Partially-Proved” means the formal engine hit its proof depth, not that it found a failure

SYNTHESIS – DC Shell:

Lib Path

```
set search_path  [list . ../src ./script
/fetools/work_area/frontend/SAED32_EDK/lib/stdcell_rvt/db_nldm]

set target_library [list saed32rvt_ss0p75v25c.db]

set link_library "* $target_library "
```

Why RVT

1 – If I use HVT I got Setup Violation Because it to Slow. (-3.64)

2 – If I use LVT I got Hold Violation Because It To Fast. (-0.06)

3 – If I use RVT I met The Timing Constraints.

SDC Constraints

```
create_clock -name Clk_Fast -period 6.6667 -waveform {0 3.3333} \
             [get_ports Clk_Fast]
create_clock -name Clk_Slow -period 14.2857 -waveform {0 7.1429} \
             [get_ports Clk_Slow]
set_clock_uncertainty -setup 0.20 [get_clocks Clk_Fast]
set_clock_uncertainty -hold  0.10 [get_clocks Clk_Fast]
set_clock_uncertainty -setup 0.40 [get_clocks Clk_Slow]
set_clock_uncertainty -hold  0.30 [get_clocks Clk_Slow]
set_clock_latency    -source 0.5 [get_clocks Clk_Fast]
set_clock_latency    -source 0.5 [get_clocks Clk_Slow]
set_clock_transition 0.10        [get_clocks Clk_Fast]
set_clock_transition 0.15        [get_clocks Clk_Slow]
set_clock_groups -asynchronous  -group [get_clocks Clk_Fast] \
                 -group [get_clocks Clk_Slow]
set_false_path -from [get_ports Rst]
set_input_delay -clock Clk_Fast -max 2.2 [get_ports {Prog_Data[*]}]
set_input_delay -clock Clk_Fast -min 1.0 [get_ports {Prog_Data[*]}]
set_input_delay -clock Clk_Slow -max 2.0 [get_ports
{Data_Rd_Data[*]}]
```

```
set_input_delay -clock Clk_Slow -min 1.0 [get_ports
{Data_Rd_Data[*]}]

set_output_delay -clock Clk_Fast -max 2.0 [get_ports {Halted
Prog_Addr[*]}]

set_output_delay -clock Clk_Fast -min 0.5 [get_ports {Halted
Prog_Addr[*]}]

set_output_delay -clock Clk_Slow -max 4.0 \

    [get_ports {Data_Wr_En Data_Rd_En Data_Addr[*] Data_Wr_Data[*]}]

set_output_delay -clock Clk_Slow -min 0.5 \

    [get_ports {Data_Wr_En Data_Rd_En Data_Addr[*] Data_Wr_Data[*]}]
```

Run.TCL

```
analyze -format verilog {

    ../Alu_Unit.v

    ../Async_FIFO.v

    ../Branch_Unit.v

    ../Cache.v

    ../Control_Unit.v

    ../Cpu.v

    ../d_ff.v

    ../Fifo_Handler.v

    ../fifo_mem.v

    ../Flag_Store.v

    ../Hazard_Unit.v

    ../Imm_Padding.v

    ../Instruction_Decode.v

    ../Mux_2x1.v

    ../Mux_4x1.v

    ../Pc_Unit.v

    ../Pipe_EX_MEM.v

    ../Pipe_ID_EX.v

    ../Pipe_IF_ID.v
```

```
    ../Pipe_MEM_WB.v
    ../rd_ptr_handler.v
    ../Register_File.v
    ../Reset_Sync.v
    ../Risc.v
    ../Stack.v
    ../Sync.v
    ../wr_ptr_handler.v
}
elaborate Risc -parameter "LINE = 8"
current_design Risc
link
source -echo Cpu.sdc
check_design > Report/check_design.rpt
check_timing > Report/check_timing_before_compile.rpt
compile_ultra
check_timing > check_timing_after_compile.rpt
write -format verilog -hierarchy \
    -output Report/Risc_netlist.v
write -format ddc -hierarchy \
    -output Report/Risc.ddc
report_clock > Report/clock.rpt
report_constraint -all_violators > Report/constraint_violations.rpt
report_timing -delay max -max_paths 20 > Report/timing_setup.rpt
report_timing -delay min -max_paths 20 > Report/timing_hold.rpt
report_area > Report/area.rpt
report_power > Report/power.rpt
report_qor > Report/Qor.rpt
```

Area_Report

```
dc_shell> report_area

****************************************
Report : area
Design : Risc_LINE8
Version: S-2021.06-SP4
Date   : Thu Oct  8 12:54:13 2026
****************************************

Library(s) Used:

    saed32rvt_ss0p75v25c (File: /fetools/work_area/frontend/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_ss0p75v25c.db)

Number of ports:                          532
Number of nets:                         12339
Number of cells:                        11452
Number of combinational cells:           8855
Number of sequential cells:              2594
Number of macros/black boxes:               0
Number of buf/inv:                        889
Number of references:                      30

Combinational area:              22355.268627
Buf/Inv area:                     1294.863685
Noncombinational area:           18464.578496
Macro/Black Box area:                0.000000
Net Interconnect area:           20429.311848

Total cell area:                 40819.847123
Total area:                      61249.158970
1
dc_shell>
```

Qor_Report

```
dc_shell> report_qor

****************************************
Report : qor
Design : Risc_LINE8
Version: S-2021.06-SP4
Date   : Thu Oct  8 13:01:01 2026
****************************************


  Timing Path Group 'Clk_Fast'
  -----------------------------------
  Levels of Logic:               24.00
  Critical Path Length:           6.31
  Critical Path Slack:            0.00
  Critical Path Clk Period:       6.67
  Total Negative Slack:           0.00
  No. of Violating Paths:         0.00
  Worst Hold Violation:           0.00
  Total Hold Violation:           0.00
  No. of Hold Violations:         0.00
  -----------------------------------

  Timing Path Group 'Clk_Slow'
  -----------------------------------
  Levels of Logic:               25.00
  Critical Path Length:           3.72
  Critical Path Slack:            6.17
  Critical Path Clk Period:      14.29
  Total Negative Slack:           0.00
  No. of Violating Paths:         0.00
  Worst Hold Violation:           0.00
  Total Hold Violation:           0.00
  No. of Hold Violations:         0.00
  -----------------------------------


  Cell Count
  -----------------------------------
  Hierarchical Cell Count:           3
  Hierarchical Port Count:         404
  Leaf Cell Count:               11449
  Buf/Inv Cell Count:              889
  Buf Cell Count:                   75
  Inv Cell Count:                  814
  CT Buf/Inv Cell Count:             0
  Combinational Cell Count:       8855
  Sequential Cell Count:          2594
  Macro Count:                       0
  -----------------------------------
```

```
  Area
  -----------------------------------
  Combinational Area:    22355.268627
  Noncombinational Area: 18464.578496
  Buf/Inv Area:           1294.863685
  Total Buffer Area:           205.86
  Total Inverter Area:        1089.01
  Macro/Black Box Area:          0.000000
  Net Area:              20429.311848
  -----------------------------------
  Cell Area:             40819.847123
  Design Area:           61249.158970


  Design Rules
  -----------------------------------
  Total Number of Nets:         11937
  Nets With Violations:             3
  Max Trans Violations:             0
  Max Cap Violations:               3
  -----------------------------------


  Hostname: feserver.mirafra.com

  Compile CPU Statistics
  ---------------------------------------
  Resource Sharing:                  3.52
  Logic Optimization:                8.91
  Mapping Optimization:             24.27
  ---------------------------------------
  Overall Compile Time:             83.51
  Overall Compile Wall Clock Time:  85.07

  --------------------------------------------------------------------

  Design  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0

  Design (Hold)  WNS: 0.00  TNS: 0.00  Number of Violating Paths: 0

  --------------------------------------------------------------------


1
```

Report_Power

```
dc_shell> report_power

****************************************
Report : power
        -analysis_effort low
Design : Risc_LINE8
Version: S-2021.06-SP4
Date   : Thu Oct  8 13:07:49 2026
****************************************


Library(s) Used:

    saed32rvt_ss0p75v25c (File: /fetools/work_area/frontend/SAED32_EDK/lib/stdcell_rvt/db_nldm/saed32rvt_ss0p75v25c.db)


Operating Conditions: ss0p75v25c   Library: saed32rvt_ss0p75v25c
Wire Load Model Mode: enclosed

Design        Wire Load Model            Library
------------------------------------------------
Risc_LINE8             70000             saed32rvt_ss0p75v25c
Cpu_LINE8              70000             saed32rvt_ss0p75v25c
Alu_Unit               16000             saed32rvt_ss0p75v25c
Cache_LINES8           16000             saed32rvt_ss0p75v25c


Global Operating Voltage = 0.75
Power-specific unit information :
    Voltage Units = 1V
    Capacitance Units = 1.000000ff
    Time Units = 1ns
    Dynamic Power Units = 1uW    (derived from V,C,T units)
    Leakage Power Units = 1pW

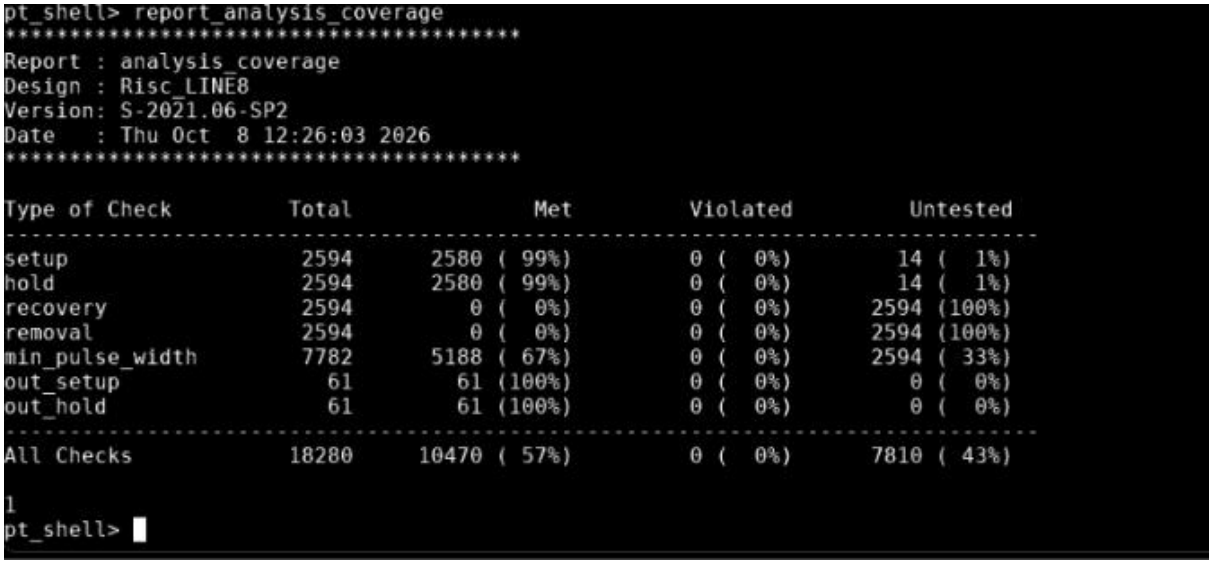

  Cell Internal Power  =   1.2881 mW   (99%)
  Net Switching Power  =  14.9740 uW    (1%)
                         ---------
Total Dynamic Power    =   1.3030 mW  (100%)

Cell Leakage Power     =  42.5310 uW


                 Internal         Switching           Leakage            Total
Power Group      Power            Power               Power              Power   (   %    )  Attrs
--------------------------------------------------------------------------------------------------
io_pad             0.0000           0.0000              0.0000             0.0000  (   0.00%)
memory             0.0000           0.0000              0.0000             0.0000  (   0.00%)
black_box          0.0000           0.0000              0.0000             0.0000  (   0.00%)
clock_network      0.0000           0.0000              0.0000             0.0000  (   0.00%)
register       1.2820e+03           1.5187          2.5625e+07         1.3092e+03  (  97.30%)
sequential         0.0000           0.0000              0.0000             0.0000  (   0.00%)
combinational      6.0203          13.4552          1.6906e+07            36.3814  (   2.70%)
--------------------------------------------------------------------------------------------------
Total          1.2881e+03 uW       14.9740 uW       4.2531e+07 pW      1.3456e+03 uW
1
```

STA – PrimeTime

- **Report_Analysis_Coverage**

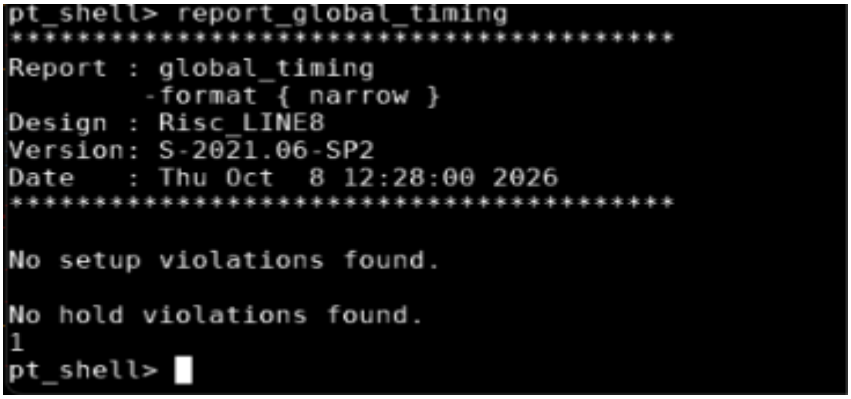


```
pt_shell> report_analysis_coverage
****************************************
Report : analysis_coverage
Design : Risc_LINE8
Version: S-2021.06-SP2
Date   : Thu Oct  8 12:26:03 2026
****************************************

Type of Check         Total              Met         Violated         Untested
-------------------------------------------------------------------------------
setup                  2594      2580 ( 99%)         0 (  0%)        14 (  1%)
hold                   2594      2580 ( 99%)         0 (  0%)        14 (  1%)
recovery               2594         0 (  0%)         0 (  0%)      2594 (100%)
removal                2594         0 (  0%)         0 (  0%)      2594 (100%)
min_pulse_width        7782      5188 ( 67%)         0 (  0%)      2594 ( 33%)
out_setup                61        61 (100%)         0 (  0%)         0 (  0%)
out_hold                 61        61 (100%)         0 (  0%)         0 (  0%)
-------------------------------------------------------------------------------
All Checks             18280     10470 ( 57%)         0 (  0%)      7810 ( 43%)

1
pt_shell>
```

- **Report_global_timing**



```
pt_shell> report_global_timing
****************************************
Report : global_timing
         -format { narrow }
Design : Risc_LINE8
Version: S-2021.06-SP2
Date   : Thu Oct  8 12:28:00 2026
****************************************

No setup violations found.

No hold violations found.
1
pt_shell>
```

- **Report_timing -delay_type max**

```
pt_shell> report_timing -delay_type max
****************************************
Report : timing
        -path_type full
        -delay_type max
        -max_paths 1
        -sort_by slack
Design : Risc_LINE8
Version: S-2021.06-SP2
Date   : Thu Oct  8 12:33:30 2026
****************************************

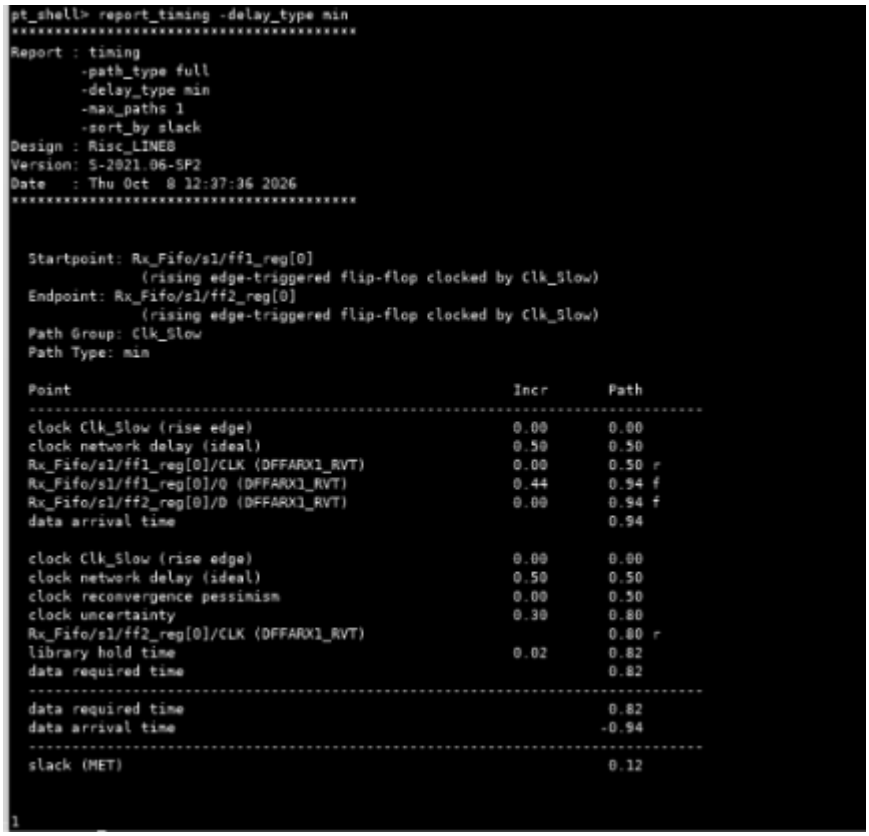

  Startpoint: Cpu/Stage_ID_EX/Pipe_Op1_reg[6]
              (rising edge-triggered flip-flop clocked by Clk_Fast)
  Endpoint: Cpu/Flag/Flags_reg[0]
              (rising edge-triggered flip-flop clocked by Clk_Fast)
  Path Group: Clk_Fast
  Path Type: max

  Point                                                    Incr       Path
  -----------------------------------------------------------------------------
  clock Clk_Fast (rise edge)                               0.00       0.00
  clock network delay (ideal)                              0.50       0.50
  Cpu/Stage_ID_EX/Pipe_Op1_reg[6]/CLK (DFFARX1_RVT)        0.00       0.50 r
  Cpu/Stage_ID_EX/Pipe_Op1_reg[6]/Q (DFFARX1_RVT)          0.60       1.10 r
  Cpu/Alu/U677/Y (XNOR2X1_RVT)                             0.41       1.51 f
  Cpu/Alu/U97/Y (INVX2_RVT)                                0.18       1.70 r
  Cpu/Alu/U61/Y (NAND2X2_RVT)                              0.33       2.02 f
  Cpu/Alu/U174/Y (OAI22X1_RVT)                             0.31       2.33 r
  Cpu/Alu/U925/CO (FADDX1_RVT)                             0.24       2.57 r
  Cpu/Alu/U945/S (FADDX1_RVT)                              0.38       2.95 f
  Cpu/Alu/U955/CO (FADDX1_RVT)                             0.26       3.21 f
  Cpu/Alu/U961/S (FADDX1_RVT)                              0.40       3.61 r
  Cpu/Alu/U959/S (FADDX1_RVT)                              0.36       3.97 f
  Cpu/Alu/U957/Y (OR2X1_RVT)                               0.17       4.14 f
  Cpu/Alu/U1206/Y (AOI21X1_RVT)                            0.27       4.41 r
  Cpu/Alu/U109/Y (OAI21X1_RVT)                             0.28       4.69 f
  Cpu/Alu/U1210/Y (NAND2X0_RVT)                            0.11       4.80 r
  Cpu/Alu/U1213/Y (AND2X1_RVT)                             0.13       4.93 r
  Cpu/Alu/U1214/Y (OA21X1_RVT)                             0.14       5.08 r
  Cpu/Alu/U29/Y (INVX1_RVT)                                0.08       5.16 f
  Cpu/Alu/U1886/Y (AOI21X1_RVT)                            0.28       5.44 r
  Cpu/Alu/U64/Y (XNOR2X2_RVT)                              0.22       5.66 f
  Cpu/Alu/U480/Y (AND4X1_RVT)                              0.23       5.89 f
  Cpu/Alu/U1918/Y (AND4X1_RVT)                             0.27       6.16 f
  Cpu/Alu/U1919/Y (OR3X1_RVT)                              0.20       6.36 f
  Cpu/Alu/U2468/Y (NOR2X0_RVT)                             0.22       6.58 r
  Cpu/Alu/U2923/Y (NAND3X0_RVT)                            0.08       6.66 f
  Cpu/U212/Y (A022X1_RVT)                                  0.16       6.81 f
  Cpu/Flag/Flags_reg[0]/D (DFFARX1_RVT)                    0.00       6.81 f
  data arrival time                                                   6.81

  clock Clk_Fast (rise edge)                               6.67       6.67
  clock network delay (ideal)                              0.50       7.17
  clock reconvergence pessimism                            0.00       7.17
  clock uncertainty                                       -0.20       6.97
  Cpu/Flag/Flags_reg[0]/CLK (DFFARX1_RVT)                             6.97 r
  library setup time                                      -0.15       6.82
  data required time                                                  6.82
  -----------------------------------------------------------------------------
  data required time                                                  6.82
  data arrival time                                                  -6.81
  -----------------------------------------------------------------------------
  slack (MET)                                                         0.00


1
pt_shell>
```

- **Report_timing -delay_type min**



```
pt_shell> report_timing -delay_type min
****************************************
Report : timing
        -path_type full
        -delay_type min
        -max_paths 1
        -sort_by slack
Design : Risc_LINE8
Version: S-2021.06-SP2
Date   : Thu Oct  8 12:37:36 2026
****************************************


  Startpoint: Rx_Fifo/s1/ff1_reg[0]
              (rising edge-triggered flip-flop clocked by Clk_Slow)
  Endpoint: Rx_Fifo/s1/ff2_reg[0]
              (rising edge-triggered flip-flop clocked by Clk_Slow)
  Path Group: Clk_Slow
  Path Type: min

  Point                                                   Incr       Path
  -----------------------------------------------------------------------------
  clock Clk_Slow (rise edge)                              0.00       0.00
  clock network delay (ideal)                             0.50       0.50
  Rx_Fifo/s1/ff1_reg[0]/CLK (DFFARX1_RVT)                 0.00       0.50 r
  Rx_Fifo/s1/ff1_reg[0]/Q (DFFARX1_RVT)                   0.44       0.94 f
  Rx_Fifo/s1/ff2_reg[0]/D (DFFARX1_RVT)                   0.00       0.94 f
  data arrival time                                                  0.94

  clock Clk_Slow (rise edge)                              0.00       0.00
  clock network delay (ideal)                             0.50       0.50
  clock reconvergence pessimism                           0.00       0.50
  clock uncertainty                                       0.30       0.80
  Rx_Fifo/s1/ff2_reg[0]/CLK (DFFARX1_RVT)                            0.80 r
  library hold time                                       0.02       0.82
  data required time                                                 0.82
  -----------------------------------------------------------------------------
  data required time                                                 0.82
  data arrival time                                                 -0.94
  -----------------------------------------------------------------------------
  slack (MET)                                                        0.12


1
```

LEC – RTL vs Netlist

Port Match:

```
Formality (setup)> match
Reference design is 'r:/WORK/Risc'
Implementation design is 'i:/WORK/Risc_LINE8'
Status:  Checking designs...

Warning: 0 (2) undriven nets found in reference (implementation) design; see formality.log for list (FM-399)
Status:  Building verification models...
Status:  Matching...

********************************** Matching Results **********************************
 2655 Compare points matched by name
 0 Compare points matched by signature analysis
 0 Compare points matched by topology
 67 Matched primary inputs, black-box outputs
 0(0) Unmatched reference(implementation) compare points
 0(0) Unmatched reference(implementation) primary inputs, black-box outputs
**************************************************************************************

1
```

Verify:

```
Formality (match)> verify
Reference design is 'r:/WORK/Risc'
Implementation design is 'i:/WORK/Risc_LINE8'

********************************** Matching Results **********************************
 2655 Compare points matched by name
 0 Compare points matched by signature analysis
 0 Compare points matched by topology
 67 Matched primary inputs, black-box outputs
 0(0) Unmatched reference(implementation) compare points
 0(0) Unmatched reference(implementation) primary inputs, black-box outputs
**************************************************************************************

Status:  Verifying...

Status:  Matching hierarchy...


Status:  Verifying...


******************************** Verification Results ********************************
Verification SUCCEEDED
----------------------
 Reference design: r:/WORK/Risc
 Implementation design: i:/WORK/Risc_LINE8
 2655 Passing compare points
--------------------------------------------------------------------------------------
Matched Compare Points     BBPin    Loop   BBNet     Cut    Port     DFF     LAT   TOTAL
--------------------------------------------------------------------------------------
Passing (equivalent)           0       0       0       0      61    2594       0    2655
Failing (not equivalent)       0       0       0       0       0       0       0       0
**************************************************************************************
Info:  Formality Guide Files (SVF) might improve the performance of this verification.
1
```